

CS630 - ASSIGNMENT 5

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1Ans.

(1.1)

Hash functions:

$$H1(X) = x \bmod 11$$

$$h2(x) = 10x-1 \bmod 11$$

$$h3(x) = [x/11] \bmod 11$$

For $x = 10, 123, 2, 0, 111, 1000$ we find the values for the hash functions

For $x = 10$

$$H1(10) = 10 \bmod 11 = 10$$

$$H2(10) = 100-1 \bmod 11 = 0$$

$$H3(10) = 10/11 \bmod 11 = 0$$

So, here we mark 1 for 0th and 10th index.

For $x = 123$

$$H1(123) = 123 \bmod 11 = 2$$

$$H2(123) = 1230-1 \bmod 11 = 8$$

$$H3(123) = 123/11 \bmod 11 = 0$$

Here, we will mark 1 for 0th, 2th and 8th indexes in the next step.

For $x = 2$

$$H1(2) = 2 \bmod 11 = 2$$

$$H2(2) = 20-1 \bmod 11 = 8$$

$$H3(2) = 2/11 \bmod 11 = 0$$

Here, we will mark for 2th, 8th and 0th as well

For $x = 0$

$$H1(0) = 0 \bmod 11 = 0$$

$$H2(0) = 0-1 \bmod 11 = 10$$

$$H3(0) = 0/11 \bmod 11 = 0$$

For $x = 111$

$$H1(111) = 111 \bmod 11 = 1$$

$$H2(111) = 1110-1 \bmod 11 = 9$$

$$H3(111) = 111/11 \bmod 11 = 10$$

For $x = 1000$

$$H1(1000) = 1000 \bmod 11 = 10$$

$$H2(1000) = 10000-1 \bmod 11 = 0$$

$$H3(1000) = 1000/11 \bmod 11 = 2$$

Here, we are marking the 0, 2 and 10 as 1

For any $h1(x)$, $h2(x)$, $h3(x)$ values if the their index value in the table is 1 then they are present, otherwise not.

INDEX	0	1	2	3	4	5	6	7	8	9	10
INITIAL	0	0	0	0	0	0	0	0	0	0	0
x = 10	1	0	0	0	0	0	0	0	0	0	1
x = 123	1	0	1	0	0	0	0	0	1	0	1
x = 2	1	0	1	0	0	0	0	0	1	0	1
x = 0	1	0	1	0	0	0	0	0	1	0	1
x = 111	1	1	1	0	0	0	0	0	1	1	1
x = 1000	1	1	1	0	0	0	0	0	1	1	1

(1.2)

We have to query if 37, 7, 1100, 124 are present or not.

So, for x = 37, 7, 1100, 124

X= 37

$$H1(37) = 37 \bmod 11 = 4$$

$$H2(37) = 370-1 \bmod 11 = 6$$

$$H3(37) = 37/11 \bmod 11 = 3$$

For none of the above hash values, its index has 1. So, 37 is negative.

X= 7

$$H1(7) = 7 \bmod 11 = 7$$

$$H2(7) = 70-1 \bmod 11 = 3$$

$$H3(7) = 7/11 \bmod 11 = 0$$

For 7 and 3 the index is 0, so 7 is not present

For x = 7 it is negative

X=1100

$$H1(1100) = 1100 \bmod 11 = 0$$

$$H2(1100) = 11000-1 \bmod 11 = 10$$

$$H3(1100) = 1100/11 \bmod 11 = 1$$

For all the 0,1 and 10, its indexes value is 1. So, 1100 might be present. Its positive

X= 124

$$H1(124) = 124 \bmod 11 = 3$$

$$H2(124) = 1240-1 \bmod 11 = 7$$

$$H3(124) = 124/11 \bmod 11 = 0$$

For 3 and 7 the values are not 1. So, 124 is not present. Its negative

(1.3)

Probability of having a number that returns positive but still not present.

Two answers are possible in the bloom filter one is negative and probably positive. So chances of false positive are

$$P = [1 - (1 - 1/m)^{kn}]^k$$

M = size of the array = 11

N = number of elements for holding the 1 = 6

K = no. of hash function = 3

$$P = [1 - (1 - 1/11)^{18}]^3$$

$$P = [1 - 0.179]^3 = 0.820^3 = 0.551$$

So, the chances of the positive values being wrong is 0.551.

2Ans

2.1.a

Bloom filters are used to check if a value is present or not. Now, we have two blooming filter A and B where both of them have same number of bits. Let the number of bits be n.

X are the elements of A and Y are the elements of B. Initially all the bits are equal to 0 for both A and B.

Let elements of X= x1, x2, x3 and y= y1, y2, where x2=y1. And hashing the values of X and Y the A and B value be like this. So, $C = A \wedge B$. where if both values of A and B are 1 the resulting value will be 1, otherwise 0. For bitwise 'and'

index	0	1	2	3	4	5
Initially	0	0	0	0	0	0
A	1	1	0	0	1	0
B	1	1	0	0	0	0
C	1	1	0	0	0	0

For $X \cap Y$ the elements will be x2.

Then, the resulting table will be

$x \cap y$	0	1	1	0	0	0
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Which is clearly not the same as C

So, C is not the same as $X \cap Y$.

2.1.b

No, C does not represent the set $X \cap Y$. so, the bits in both the tables are not the same. If there is a query then the probability of getting a positive answer is not sure. Because, the values get filtered in the intersection. C is a bitwise AND, which means only if it's 1 for both then the resulting value is 1. For most of the query the result might be negative.

2.1.c

Let us take the same example as above. The D= bitwise A OR B. That means it is 1 if either of the bit is 1 .

A	1	1	0	0	1	0
B	1	1	0	0	0	0
C	1	1	0	0	1	0

$X \cup Y = x1, y1, x3, y2$. The resulting table will be

$X \cup Y$	1	1	0	0	1	0
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It is true, D represents the union of X and Y.

2.2

Here we are dividing the number of bits of B into two halves. The total number of bits on B is m. So, the first half be B1 and the other be B2. $B' = B1 \vee B2$.

$h_1(x) \dots h_r(x)$ are hash functions used on B. So, as in the 2.1.c problem the bits formed by union of two elements is the same as the one formed by its original B formed by $B_1 \vee B_2$. By computing the values in B' with $i \text{ hi}(x) \bmod (m/2)$ will be the same as B. So, for the query of the element belongs to B' will return positive is it return positive for B.

3Ans

There are n socks. Consider the number of red socks are 'r' and number of blue socks will be 'n-r'. So, the probability of getting the red socks is r/n and probability of getting the blue socks will be $(n-r)/r$.

3.1.a

We are using the Markov chain model. If there are $n+1$ states then S is the state. The s_i corresponds to the number of the red socks. Therefore, for S_0 shows 0 red socks, for s_1 there is 1 red socks, S_2 there is 2 red socks and so on. When we draw a socks and it is red, we throw it away and replace it with the blue one. The number of blue socks inside increases and the state for red socks decreases. So, if we replace the red with blue the state becomes S_{i-1} . And when we get the blue sock out the drawer, we keep it back inside. So, the state changes from S_{i+1} . So, the state transitions to S_{i+1} . The probability of state transition here depends on the number of red socks inside. It transitions until all the red socks are thrown out.

3.1.b

The Markov chain property is memoryless. We can say this by observing that the probability of one state to other transition is dependent on the number of red socks that are there. But, it is always S_{i+1} means it is not depending on the past values or the color that we got. In the present, it considers the value we got and calculates the probability for the next step.

3.1.c

P_{ij} is a probability between s_i and s_j . The states are $s_i = s_{i-1}$ and $s_j = s_{i+1}$. So, if we are drawing out a red sock, then the state changes to s_{i-1} . The probability of transitioning to s_{i-1} is probability of drawing a red sock is $(n-i)/n$. If we get a blue sock out of the drawer then the state changes to s_{i+1} which is s_j . So, transitioning to s_j is probability of getting a blue sock i/n . The p_{ij} is $P(s_i \text{ to } s_j)$.

3.1.d

This Markov chain is irreducible and aperiodic. Because, while transition from one state to other state it depends on the present value and doesn't remember the past step, and we can transition to any other state, so it is irreducible. It is aperiodic, from one state to any state or to its own, it's value is greater than 0. So, the probability of chance of the state to any other state is more than 0. Therefore, the transition from one state to other doesn't not depend on the colour of the socks that we got earlier and probability of getting the next sock as blue or red is greater than 1.

3.1.e

The Markov chain has a stationery distribution. The transition probabilities are equal, the probability of moving from the initial to the current state is the same.

3.2

Here, if we get the blue socks twice in a row then we are removing it.

3.2.a

True, the Markov chain is memory less. The probability of moving from one state to other depends on the current possibility only. That means, it does not remember the previous state and calculate the probability. It is not considering any of the past values. The probability just depends on the present state.

3.2.b

The probability of getting a red socks with the condition that the previous socks was blue, which is getting aa red sock only. That is $n/n+1$. The probability of getting a blue socks with the condition that the before one was also blue is subtracting the probability of red from 1(Total chances).

So, $P(r/b) = p(r) = n/n+1$
And the $P(b/b) = 1 - p(r/b)$
 $= 1 - (n/n+1).$

3.2.c

True, this Markov chain is irreducible and aperiodic. The transition from one to the other state is possible. The probabilities from one to other state is not zero. Means, that the probability of getting a red, blue or blue twice in a row is possible in every state. So, it is irreducible and aperiodic.

4Ans.

4.1

We know $m(\text{edge})$ value and $n(\text{vertices})$ is unknown. The sampler picks a random edge (u,v) and return $\text{degree}(u)$. The $\text{degree}(u)$ denotes the degree of vertex u .

Probability of returning $\text{degree}(u)$ is $1/2$. So, is each vertex if there are two edges, $2m$ is the maximum number of edges we can have. The degree of a vertex (u) is the number of edges that are on it.

$P(u) = \text{no. of edges on } u / \text{Total no. of edges}$
 $= \text{deg}(u)/2m.$

4.2

The expected random variable will be product of sum of values of F and its probabilities.

$E(x) = \sum (P(u) \times f(u))$
 $= \sum (2 \text{ deg}(u)^2)$ [2m is the maximum number of edges we can have]
 $= 2 \sum (\text{deg}(u))^2 / m$
 $= 8m$